

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
10/1	(a)			Dave: Long distance cf middle distance: {4 is not double 2.5 / 120 is not double 80 / 30 is not double 20 / double 2.5 = 5 / double 80 = 160 / double 20 = 40} (1) Tom: Olympic cf sprint: {1.5 km is double 750 m / 40 is double 20 / 10 is double 5} (1) (So Dave is wrong / Tom is correct)			2	2	2	
	(b)			Airflow/minute at 4 minutes = $13.1 \times 2.9 = 37.99/38$ (1) Airflow/minute at 3 minutes = $13.5 \times 2.7 = 36.45/36.5$ (1) (So Petula is correct) Allow 1 mark for 13.1×2.9 AND 13.5×2.7			2	2	2	
	(c)	(i)		$76/1.7^2$ (1) = 26 / 26.3 / 26.2965 (1) Allow 26.2 / 26.29 for 1 mark only		2		2	2	
		(ii)		$220 - 23 = 197$		1		1	1	
	(d)			114 to 152			1	1		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(e)			Olympic triathlon (1) + Any 2 × (1) from: swim time of (just over) 30 min / swim distance of ~1.5 km (1) cycle time of 1.3 h / cycle distance of ~40 km (1) run time of (just under/about) 1h / run distance of ~10km (1) Alternatives for the remaining two marks: Graph shows a total time of 2.74 h (1) and the closest triathlon time is 2.88 h (1) OR Graph shows a total distance of 51.5 km (1) the total distance in table 1 is $1.5 + 40 + 10 = 51.5$ (1) OR Graph shows total time of 2.74 (1) + transition time of 0.14 = 2.88 (1)			3	3	3	
	(f)			Distance = $3.8 + 180 + 42.2 = 226$ km (1) Mean speed = 226 (ecf from addition shown above) / 12.8 (1) = $17.66/17.7$ (ecf) km/h (1) For 2 marks: 17.6 / 17.65 on answer line		3		3	3	
	(g)			Change in speed = $6.2 - 1.5 = 4.7$ (1) Acc = 4.7 (ecf from subtraction shown above) / 26 (1) = $0.18 / 0.2$ (1)		3		3	3	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(h)			From 18-24 to 35-39 the mean speed increases (1) From 35-39 upwards the mean speed decreases (for every age group) (1) So true after 39 but not before The peak speed is for 35-39 yr-olds (1) People below 35 and above 39 are slower (1) So only half true Conclusion required for 2 marks No mark for comparison of 18-24 with 70-74 without intermediate age range of 35-39			2	2		
				Question 10/1 total		9	10	19	16	